

DPL30 Diaphragm Pump

Specification sheet



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1 Appearance and Dimensions

1.1 DPL30-24(12)DB-EP/PS

Unit: mm

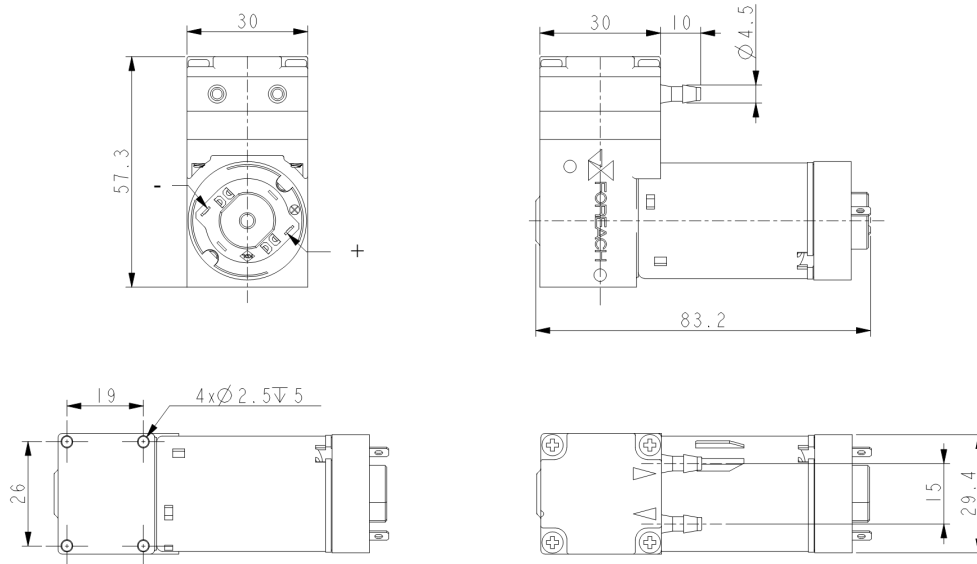


Figure 1 DPL30-24(12)DB - EP/PS Outline Dimensions

1.2 DPL30-24(12)BB-EP/PS

Unit:mm

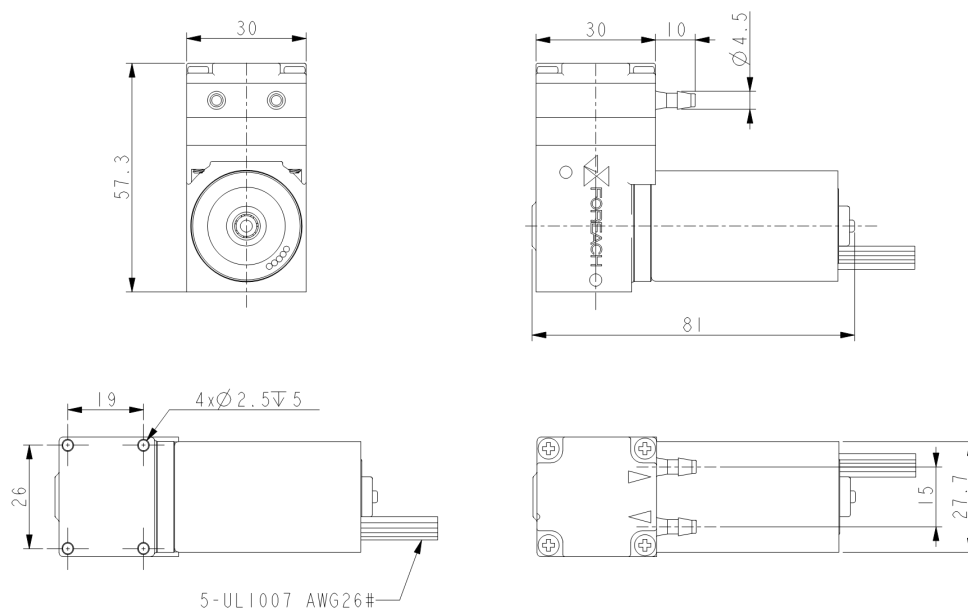


Figure 2 DPL30-24(12)BB - EP/PS Outline Dimensions

2 Specifications

Table 1 Electrical Specifications

Electrical Specifications	
Label	Function Description
+	Positive Terminal
None	Negative Terminal

Table 2 Wire Functions

Wire Functions		
Wiring	Color	Function Description
VCC	Red	Positive Power Supply
GND	Black	Ground Wire
F/R	Blue	Direction Control: Hi-Z/Open = Clockwise (CW); GND = Counter-Clockwise (CCW)
FG	Green	Tachometer Output (FG)
PWM	White	When the wiring is open or connected to 0~0.25V, the motor stops. When connected to 4.5V~5V, the motor runs at full speed. The speed can be adjusted by PWM or 0~5V connection, with the motor speed being directly proportional to the PWM duty cycle.

Note:

The information in this specification is based on nominal unit technical data and test results. It is considered accurate and reliable, and is provided as an aid in selecting Foreach products. Customers are responsible for determining product suitability for their intended applications and for conducting necessary tests prior to use.

Table 3 Technical Specifications

Technical Specifications				
Specification	Parameter			
Model	DPL30-24DB-EP/PS	DPL30-12DB-EP/PS	DPL30-24BB-EP/PS	DPL30-12BB-EP/PS
Motor Type	Brushed DC	Brushed DC	Brushless DC	Brushless DC
Rated Voltage	DC24V±10%	DC12V±10%	DC24V±10%	DC12V±10%
Power	≤8W			
No-Load Flow Rate	300mL/min			
Rated Pressure	100KPa			
Self-Priming Height	5m			
Working Medium	Purified Water (Other liquid medium needs to be evaluated by the customer.)			
Medium Temperature Range	+5℃~+40℃			
Weight	160g		185g	
Tubing Connection	Ø3.2mm ID Hose			
Lifespan	3000h (Rated Voltage, Continuous Operation)		10000h (Rated Voltage, Continuous Operation)	
Noise	≤80dB			
IP Rating	00		00	
Operating Temperature	+5℃~+40℃			
Operating Humidity	30%-85% RH			
Storage Temperature and Humidity	-10℃~+40℃ 10%-90% RH			

Table 4 Material Options

Material Options	
Components	Material
Pump Head	PPS
Diaphragm	EPDM/PTFE
Valves	EPDM/FFKM

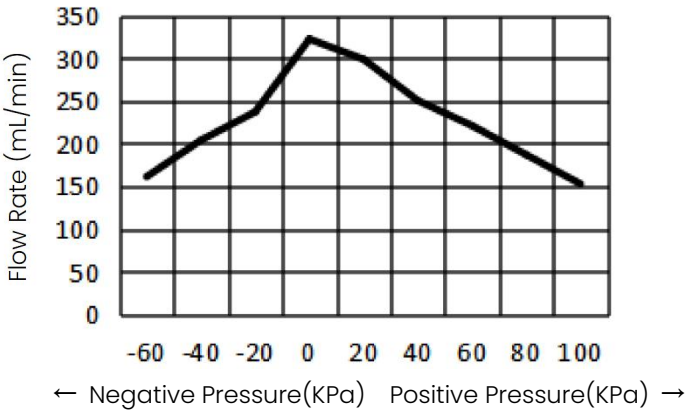


Figure 3 Flow Rate - Pressure Curve

3 Ordering Information

Table 5 Order List

Product Model Information List						
No.	Product Code	Product model	Motor Type	Rated Voltage	Diaphragm	Valve
1	459001	DPL30-24DB-EP/PS	Brushed DC	24V	EPDM	EPDM
2	459013	DPL30-12DB-EP/PS	Brushed DC	12V	EPDM	EPDM
3	459002	DPL30-24BB-EP/PS	Brushless DC	24V	EPDM	EPDM
4	459014	DPL30-12BB-EP/PS	Brushless DC	12V	EPDM	EPDM
5	459027	DPL30-12DB-FF/PS	Brushed DC	12V	PTFE	FFKM
6	459025	DPL30-24DB-FF/PS	Brushed DC	24V	PTFE	FFKM
7	459028	DPL30-12BB-FF/PS	Brushless DC	12V	PTFE	FFKM
8	459026	DPL30-24BB-FF/PS	Brushless DC	24V	PTFE	FFKM

DPL30	- 24	D	B	EP/PS	- X
Product Series	Operating Voltage	Motor Type	Connection Type	Diaphragm/ Housing material	Special Customization
	24: DC24V	D: Brushed DC	B: Barbed Port	EP/PS: EPDM/PPS	X: Custom Code
	12: DC12V	B: Brushless DC	S: Threaded Port	FF/PS: PTFE+FFKM/PPS	
	5: DC5V	C: Coreless Motor			

Figure 4 DPL30 Selection Table

For example:

DPL30-24DB-EP/PS: Diaphragm liquid pump flow rate 300ml/min, rated voltage 24V brushed DC motor, barbed tubing connection, diaphragm material EPDM, valve material EPDM and housing material PPS.

4 Appendices

4.1 Appendix : Tables and Figures

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DPL30-24(12)DB - EP/PS Outline Dimensions
Figure 2
DPL30-24(12)BB - EP/PS Outline Dimensions
Figure 3
Flow Rate - Pressure Curve
Figure 4
DPL30 Selection Table

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